

Problem 1:

1 2 & 7

1 2₁₀ i n B a s e - 2:

1 2 ÷ 2 = 6 , R e m = 0

6 ÷ 2 = 3 , R e m = 0

3 ÷ 2 = 1 , R e m = 1

1 ÷ 2 = 0 , R e m = 1

1 2 = 1 1 0 0

7₁₀ i n B a s e - 2 :

7 ÷ 2 = 3 , R e m = 1

3 ÷ 2 = 1 , R e m = 1

1 ÷ 2 = 0 , R e m = 1

7₁₀ = 1 1 1₂ = 0 1 1 1₂

1 1 0 0
0 1 0 0

(0100)₂ = 4₁₀

Write in decimal as
the question is in decimals

Eg: Convert
243₁₀ to boolean

2 | 243

2 | 121 - 1

2 | 60 - 1

2 | 30 - 0

2 | 15 - 0

2 | 7 - 1

2 | 3 - 1

1 = 1

243₁₀ = (11110011)₂

Problem 2: (R: remainder; Q: quotient)

$$15 \mid 10$$

15_{10} in Base - 2 :

$$15 \div 2 \quad Q = 7 \quad R = 1$$

$$7 \div 2 \quad Q = 3 \quad R = 1$$

$$3 \div 2 \quad Q = 1 \quad R = 1$$

$$1 \div 2 \quad Q = 0 \quad R = 1$$

$$15_{10} = 1111_2$$

10_{10} in Base - 2 :

$$10 \div 2 \quad Q = 5 \quad R = 0$$

$$5 \div 2 \quad Q = 2 \quad R = 1$$

$$1 \div 2 \quad Q = 0 \quad R = 1$$

$$10_{10} = 1101_2$$

$$\begin{array}{r} 111 \\ 111 \\ \hline 111 \end{array}$$

X

1111

Problem 3-20¹³:

20_{10} in Base - 2:

$$20 \div 2 Q = 10, R = 0$$

$$10 \div 2 Q = 5, R = 0$$

$$5 \div 2 Q = 2, R = 1$$

$$2 \div 2 Q = 1, R = 0$$

$$1 \div 2 Q = 0, R = 1$$

20_{10} in Base - 2 is 10100_2

13_{10} in Base - 2:

$$13 \div 2 Q = 6, R = 1$$

$$6 \div 2 Q = 3, R = 0$$

$$3 \div 2 Q = 1, R = 1$$

$$1 \div 2 Q = 0, R = 1$$

13_{10} in Base - 2 is $1101 = 01101$

$$\begin{array}{r} 10110 \\ 01101 \\ \hline 11011 \end{array} \quad \times$$

$$\begin{array}{r} 10100 \\ 01101 \\ \hline (11001)_2 = 25_{10} \end{array}$$

Problem 4 - ~10:

10₁₀ in Base - 2:

$$10 \div 2 \quad Q = 5, \quad R = 0$$

$$5 \div 2 \quad Q = 2, \quad R = 1$$

$$2 \div 2 \quad Q = 1, \quad R = 0$$

$$1 \div 2 \quad Q = 0, \quad R = 1$$

1010

0101

$$\sim(10) = \sim(1010) = (0101)_2 = 5_{10}$$

Python uses sign - magnitude format for negative numbers, while 2s complement method is used at the hardware level to store negative integers.

See
AI notes

for this
problem

To represent -ve numbers, you can use first bit
to represent sign if you 8-bit numbers. 1 → +ve
0 → -ve
Remaining 7-bits represent 0 to 2⁷-1, 0 to 127
Depending on first bit, it's -127 to 0 to +127.

$$10_{10} = (10001010)$$

$$\sim(10)_{10} = \begin{matrix} \text{sign bit} \\ \downarrow \\ (01110101) \end{matrix} = -117_{10}$$

-ve

In python, first bit 0 → +ve
first bit 1 → -ve but you
add the largest power in -ve
10₁₀ = (00001010)
~(10) = ^{sign}(11110101) = -128 + 64 + 32 + 16 + 4 + 1 = -117

Problem 5-8 < 2:

8₁₀ in Base - 2:

$$\begin{array}{r} 2 \overline{) 8} \\ \underline{4} \\ 2 \\ \underline{2} \\ 0 \\ \underline{0} \\ 0 \end{array}$$

$$8_{10} = 1000_2$$

$$1000_2 < < 2 = \begin{array}{c} \boxed{1000} \\ 0000 \\ 0000 \end{array}$$

$$8_{10} < < 2 = 1000_2 = 100000_2$$

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Problem 6-64₁₀ >> 3:

64₁₀ in Base-2:

2	6	4	
2	3	2	0
2	1	6	0
2	8		0
2	4		0
2	2		0
2	1		0
2	0		1

$$64_{10} = 1000000_2$$

$$\begin{array}{ccccccc} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{array}$$

$$64_{10} < 3 = 1000000_2 > 73 = 000100_2$$

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Problem 7 - $(5_{10} \& 3_{10}) \div 4_{10}$:

5_{10} in Base-2 :

$$\begin{array}{r} 2 \overline{) 5} \\ 2 \overline{) 2} \quad \underline{-1} \\ 2 \overline{) 1} \quad \underline{-0} \\ 2 \overline{) 0} \quad \underline{-1} \end{array}$$

$5_{10} = 101_2$

3_{10} in Base-2 :

$$\begin{array}{r} 2 \overline{) 3} \\ 2 \overline{) 1} \quad \underline{-1} \\ 2 \overline{) 0} \quad \underline{-1} \end{array}$$

$3_{10} = 11_2 = 011_2$

4_{10} in Base-2

$$\begin{array}{r} 2 \overline{) 4} \\ 2 \overline{) 2} \quad \underline{-0} \\ 2 \overline{) 1} \quad \underline{-0} \\ 2 \overline{) 0} \quad \underline{-1} \end{array}$$

$4_{10} = 100_2$

$$\begin{array}{r} 101 \\ 011 \\ \hline 001 \end{array}$$

$$\begin{array}{r} 001 \\ 100 \\ \hline 101 \end{array}$$

↓
= 5

problem 8: $(10_{10} \gg 1) \& 7_{10}$:

10₁₀ in Base-2:

2	10	
2	5	
2	2	1
2	1	0
2	0	1

10₁₀ = 1000₂ 1010₂

7₁₀ in Base-2:

2	7	
2	3	1
2	1	1
2	0	1

7₁₀ = 111₂

→

1	0	1	0
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0101

0101 = 101
&
101
101
101

$(1010 \gg 1) = (0101)$

&
111

0101 = 5₁₀

Problem 9-1¹1:

$$1 \quad / \quad / \quad 2 \quad R = 1$$

$$\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \quad \begin{array}{c} 0 \\ 1 \\ \hline 1 \end{array}$$

Problem 10 - When I do bin(-5) in Python, it gives
-0b10101

(From problem 4)

$$\sim(-5)$$

$$= \sim(1111011)$$

$$= 0000100$$

$$= 4$$

$$\begin{aligned} -5_{10} &= (1111010+1) \\ &= (1111011)_{101} \end{aligned}$$

$$\begin{aligned} &-128 + 64 + 32 + 16 + 8 + 0 + 2 + 0 \\ &\quad + 1 \end{aligned}$$

$$\sim(-5) = -(-5) - 1$$

$$= 5 - 1$$

$$= 4$$

HW: Practice examples

Problem 11 - True or False and False:

F: False; T: True

F and F = F

T or F = T

0

True or False and False

= True or (False and False)

= True or False

= True

✓

Problem 12 - not (T and F):

T a n d F = F

n o t F = T



Problem 13 - $(5 > 3)$ and $(2 == 2)$:

$$5 > 3 = T$$

$$1 == 2 = T$$

$$T \text{ and } T = T$$

Problem 14-0 or 5:

0 or 5 = F or T
5 = T

→

Problem 15: 10 and 20:

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Problem 16- not("apple" == "orange"):

" a p p l e " == " o r a n g e " = f

n o t (f) = T



Problem 17- a1=6:

$$T \neq T = F$$

Problem 18 ^{8 bit} left circular shift on $128_{10} (10000000_2)$ by 1:

$128_{10} = 10000000_2$

Left shift $\rightarrow 10000000_2$

Circular left shift

$= 00000001_2$

$= 1_{10}$

Problem 19: $\text{bool}((10)^{\wedge} \text{bool}(10))$

$$\text{bool}(\text{bool}(10)) = \text{F}$$

$$\text{bool}(\text{bool}(\text{bool}(10))) = \text{T}$$

$$\text{T}^{\wedge} \text{F} = \text{T}$$

Problem 20: $(7_{10} \& 1_{10}) == 1_{10}$ and $(8_{10} \& 1_{10}) == 4_{10}$

$$7_{10} = 111_2 \quad (\text{From problem 8})$$

$$8_{10} = 1000_2 \quad (\text{Problem 8})$$

$$4_{10} = 100_2$$

$$\& \begin{array}{r} 111 \\ 001 \\ \hline 001 \end{array} \quad \begin{array}{r} 1000 \\ 0100 \\ \hline 0000 \end{array}$$

T

T

$$T(\text{and}) T = T$$

Problem 21—T or F and not T:

	T	or	F	and	not	T
1	T	or	(F	and	(not	T))
2	T	or	(F	and	F)	
3	T	or	F			
4	T					

problem 10 — $(23 > 42) \text{ and } (36 > 12) \text{ or } (39 < 39)$:-

$$(23 > 42) = F$$

$$36 > 12 = T$$

$$(39 < 39) = T$$

$$\text{problem} = (F \text{ and } T) \text{ or } T$$

$$= F \text{ or } T$$

$$= T$$

Problem (23) $10+5 \ll$:

$$1 \quad 0 + 5 = 1 \quad 5$$

$$1 \quad 5_{10} \quad \text{in base-2} = 1 \quad 1 \quad 1 \quad 1_2 \quad (\text{Problem 2})$$

$$15 \ll 1 = (1 \quad 1 \quad 1 \quad 1_2) \ll 1$$



$$= 1 \quad 1 \quad 1 \quad 1 \quad 0_2$$

$$= (16 + 8 + 4 + 2)_{10}$$

$$= 30_{10}$$

Problem 2.4 - $A \vee (\neg A \wedge B)$:

$$\begin{aligned} & A \vee (\neg A \wedge B) = \overbrace{} \\ & (A \vee \neg A) \wedge (A \vee B) \leftarrow \\ & \text{=} T \wedge (A \vee B) \\ & \text{=} A \vee B \end{aligned}$$

Problem 25 $\neg(A \text{ and } B) \text{ or } (A \text{ and not } B)$ $\rightarrow \neg A$

$V = A \text{ and } (B \text{ or not } B)$

$= A \text{ and } T$

$= A$

Problem 26: $A \text{ or } (F \text{ and } B) \Rightarrow V$

$V = A \text{ or } F \text{ [} F \text{ and } B = F \text{]}$

$= A$

Problem 27: $\text{not}(\text{not } A \vee B) \stackrel{?}{=} V$

$$V = (\text{not}(\text{not } A)) \text{ and } (\text{not}(\text{not } B))$$

$$= A \text{ and } B$$

problem 28-5 x 2² & 15

$$2^2 = 4$$

$$5 \times 4 = 20$$

$$20_{10} = 10100_2$$

$$15_{10} = 00111_2$$

$$\begin{array}{r} 10100 \\ \underline{00111} \\ 00111 \end{array}$$

$$00100_2$$

$$= (5 \times 2^2) \& 15$$

$$= (5 \times 4) \& 15$$

$$= 20 \& 15$$

$$= 100_2$$

$$= 4_{10}$$

Problem 29 - not $(15/3 == 5 \text{ and } 7 \div 2 == 0)$

$15 / 3 == 5$ is T

$7 \div 2 == 0$ is F

T and F is F

not F is T

Problem 30 - (A or B) and C not A or B

= (A or not A) and B

= T and B

= B

Problem 31: $(A \text{ and } B \text{ and } C) \text{ or } (A \text{ and } B \text{ and not } C)$

$= A \text{ and } B \text{ and } C \text{ or not } C$
 $= A \text{ and } B \text{ and } 1$
 $= A \text{ and } B$

Problem 32 - ~5 & 7:

$$5_{10} = 101_2$$

$$\begin{array}{r} 101 \\ \hline 010 \end{array}$$

$$7_{10} = 111_2$$

$$\begin{array}{r} 010 \\ \hline 010 \end{array}$$

Problem 33— $1 \ll 3 \mid 1 \ll 1$:

$$\begin{array}{r} \\ \\ \\ \\ \end{array}$$

$$\begin{array}{r} \\ \\ \\ \end{array}$$

$$\begin{array}{r} \\ \\ \\ \\ \hline \end{array}$$

Problem 34: A and $(A \text{ or } \emptyset)$

$$A \text{ and } (A \text{ or } B) = A \text{ (Absorption)}$$

problem 35:

$T \quad o \quad r \quad F = T$

$T \quad o \quad r \quad T = T$

$n \quad o \quad t \quad (\quad T \quad) \quad = \quad F$

$T \quad a \quad n \quad d \quad F = F$

problem 36:

$A \text{ or } (A \text{ and } B) = A$

$B \text{ and } F = F$

$A \text{ or } F = A$

Problem 37:

0 0 0 0 0 0 0 1
1 0 0 0 0 0 0 0

Problem 38:

$$10 \geq 5 > 2 = 7$$

problem 39:

$$5 < 2 = F$$

$$10 / / 2 = = 5 = T$$

$$not(F) = T$$

$$T \vee T = T$$

Problem 40:

$C = A$ and B

C or not $C = T$